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**DESIGN AND IMPLEMENTATION OF A CROWDSENSED DRIVE
TEST**

MOBILE APPLICATION FOR MOBILE NETWORK COVERAGE

PREDICTION AND COVERAGE HOLE ANALYSIS

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ABSTRACT

Mobile communication networks underpin essential digital services including mobile banking, e-learning, and e-government, particularly in developing regions such as Cameroon. Ensuring acceptable Quality of Service requires continuous monitoring of radio coverage to detect coverage holes - geographic areas of poor signal quality. Traditional coverage assessment relies on physical drive testing using specialised, calibrated vehicle-mounted equipment, an approach that is accurate but expensive, time-consuming, and difficult to scale across challenging terrain and limited infrastructure typical of developing regions.

This project addresses this problem by designing and implementing a crowdsensed drive test mobile application, named DriveTest Pro, that leverages ordinary smartphones as Class-2 IoT devices to collect cellular network measurements - including Reference Signal Received Power (RSRP), Reference Signal Received Quality (RSRQ), Signal-to-Interference-plus-Noise Ratio (SINR), and geo-referenced GPS coordinates - and transmits this data to a backend analytics platform for mobile network coverage prediction and coverage hole analysis.

The work followed a structured software engineering methodology comprising six phases: requirement gathering through secondary research, reverse engineering of existing drive test tools, and contextual inquiry; requirement analysis producing a formal Software Requirements Specification (SRS) with twelve functional and ten non-functional requirements; system modelling using six UML and structured analysis diagrams (Context, Dataflow, Use Case, Sequence, Class, and Deployment diagrams); UI design covering application identity, visual design, and a fully functional frontend prototype; and database design producing a normalised relational schema implemented in both PostgreSQL with PostGIS spatial extensions and MySQL 8.0.

The resulting system was implemented as a three-tier architecture consisting of a mobile collector application (Android/iOS), a cloud-hosted backend platform performing data ingestion and coverage analysis, and a web-based analytics dashboard for RF planning engineers and network operations teams. The mobile application was demonstrated to collect RF measurements at a 1Hz sampling rate, buffer data locally during offline periods, and automatically detect coverage holes using a configurable RSRP threshold. The dashboard demonstrated real-time coverage heatmap visualisation, session management, and analytics reporting.

The key contribution of this project is a working demonstration that crowdsensed smartphone measurements can approximate the function of professional drive test equipment at a fraction of the cost, while addressing the specific operational, privacy, and trust constraints identified through structured user reluctance assessment. The project recommends further work in coverage prediction algorithm validation, expanded 5G New Radio support, and field trials comparing measurement accuracy against calibrated professional equipment.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
CQI	Channel Quality Indicator
DFD	Dataflow Diagram
DRE	Defect Removal Efficiency
EARFCN	E-UTRA Absolute Radio Frequency Channel Number
FR	Functional Requirement
GDPR	General Data Protection Regulation
GPS	Global Positioning System
IoT	Internet of Things
KPI	Key Performance Indicator
LTE	Long Term Evolution
MNO	Mobile Network Operator
MoSCoW	Must have, Should have, Could have, Won't have (prioritisation)
NFR	Non-Functional Requirement
NOC	Network Operations Centre
PCI	Physical Cell Identity
PostGIS	Spatial database extender for PostgreSQL
QoS	Quality of Service
RAT	Radio Access Technology
RF	Radio Frequency
RSRP	Reference Signal Received Power
RSRQ	Reference Signal Received Quality
RSSI	Received Signal Strength Indicator
SBC	Single Board Computer
SINR	Signal-to-Interference-plus-Noise Ratio
SQA	Software Quality Assurance
SRS	Software Requirements Specification
TLS	Transport Layer Security
UML	Unified Modelling Language
UUID	Universally Unique Identifier
WGS84	World Geodetic System 1984

CHAPTER ONE

GENERAL INTRODUCTION

1.1 Background and Context of the Study

Mobile communication networks play a critical role in enabling digital services such as voice communication, mobile broadband, e-learning, mobile banking, and e-government applications. To ensure acceptable Quality of Service (QoS) and Quality of Experience (QoE), mobile network operators (MNOs) must continuously monitor radio coverage and identify areas with poor signal quality, commonly referred to as coverage holes.

Traditionally, network coverage assessment relies on physical drive testing, where specialised equipment and calibrated devices are mounted on vehicles to collect radio frequency (RF) measurements such as Reference Signal Received Power (RSRP), Signal-to-Interference-plus-Noise Ratio (SINR), and throughput along predefined routes. A professional drive test rig typically requires a vehicle with a reinforced roof to house multiple antennas with adequate physical separation, a Single Board Computer (SBC) to coordinate voice and data measurement units, and a control laptop running specialised software. While this approach yields highly accurate, calibrated measurements, it is expensive, time-consuming, and difficult to scale, particularly in developing regions such as Cameroon, where challenging terrain, fuel-dependent vehicles on poor road infrastructure, and limited capital budgets constrain the frequency and geographic coverage of drive testing campaigns.

Recent advances in smartphone sensing capabilities, mobile operating systems, and wireless Application Programming Interfaces (APIs) have enabled ordinary smartphones to function as Class-2 IoT devices, capable of collecting geo-referenced network performance data in real time. This has given rise to mobile crowdsensing, an emerging paradigm in which measurements collected from multiple ordinary user devices complement or partially replace traditional drive tests. When combined with backend analytics and prediction models, crowdsourced data can support coverage prediction and coverage hole detection at significantly lower cost and with broader geographic reach than professional drive testing alone.

It is within this context that this project undertakes the design and implementation of a mobile crowdsensing application, named DriveTest Pro, that collects drive-test-like measurements

from smartphones and transmits them to a network monitoring platform for coverage prediction and coverage hole analysis.

1.2 Problem Statement

Mobile network operators in Cameroon and similar developing regions face a persistent operational challenge: maintaining visibility into network coverage quality across their service areas without incurring the prohibitive costs of continuous professional drive testing. Professional drive test campaigns require specialised vehicles, calibrated RF measurement hardware costing tens of thousands of dollars, trained technicians, and substantial fuel and logistics expenditure to traverse poor road networks. As a direct consequence, drive testing is typically conducted infrequently and only along a limited set of predefined routes, leaving large areas particularly in peri-urban and rural zones, without recent or any coverage data.

This data gap means that coverage holes may persist undetected for extended periods, directly degrading the Quality of Service experienced by subscribers and undermining the reliability of mobile banking, e-learning, and emergency communication services that depend on consistent connectivity. There is, therefore, a need for an alternative or complementary data collection mechanism that is significantly lower in cost, can scale to cover wider geographic areas through the participation of many ordinary users, and can still yield measurement data of sufficient quality to support meaningful coverage analysis and coverage hole detection.

This project addresses the following central problem: how can a mobile application, running on commodity smartphones already in the hands of ordinary users, be designed and implemented to collect cellular network measurements with sufficient accuracy and contextual richness to support mobile network coverage prediction and coverage hole detection, while overcoming the privacy, trust, and usability barriers that influence user willingness to adopt such an application?

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this project is to design, implement, and demonstrate a crowdsensed drive test mobile application that uses smartphones as Class-2 IoT devices to collect cellular network measurements and transmit them to a backend platform for mobile network coverage prediction and coverage hole analysis.

1.3.2 Specific Objectives

1. To identify and analyse the stakeholders, functional and non-functional requirements of a crowdsensed drive test application through structured requirement gathering techniques.
2. To produce a formal Software Requirements Specification (SRS) that documents the functional requirements, non-functional requirements, constraints, and dependency relationships of the proposed system.
3. To design the system architecture using established UML and structured analysis notations, including Context, Dataflow, Use Case, Sequence, Class, and Deployment diagrams.
4. To design and implement a relational database schema, normalised to Third Normal Form, capable of storing geo-referenced cellular network measurements and supporting spatial coverage queries.
5. To design and implement a functional mobile application user interface and an analytics dashboard that together demonstrate the complete data collection, transmission, and visualisation workflow.
6. To evaluate the resulting solution against the stated objectives and to identify its contributions, limitations, and opportunities for further work.

1.4 Proposed Methodology

This project adopts a structured, requirement-driven software engineering methodology comprising six sequential phases, each producing a defined set of deliverables that inform the subsequent phase:

1. Requirement Gathering: stakeholder identification, and data collection through secondary research (academic literature, industry documentation), reverse engineering of existing drive test tools (TEMS Pocket, G-NetTrack, NetMonster), and contextual inquiry into professional drive test operations.
2. Requirement Analysis: review of gathered requirements for completeness, consistency, and feasibility; classification into functional and non-functional requirements; prioritisation using the MoSCoW framework; and production of a formal Software Requirements Specification.

3. System Modelling and Design: production of six diagrams (Context, Dataflow, Use Case, Sequence, Class, and Deployment) to formally model the system from complementary structural and behavioural perspectives.
4. UI Design and Implementation: definition of application identity (branding, colour system, typography), visual design of mobile and dashboard screens, and implementation of a functional frontend prototype.
5. Database Design and Implementation: conceptual and logical data modelling, production of an Entity Relationship Diagram, and implementation of the schema in both PostgreSQL (with PostGIS spatial extensions) and MySQL.
6. Evaluation: assessment of the implemented solution against the stated functional and non-functional requirements.

1.5 Research Questions

1. What are the functional and non-functional requirements of a smartphone-based crowdsensed drive test application, as derived from stakeholder needs and operational constraints of traditional drive testing?
2. What system architecture and data model can support reliable collection, offline buffering, and transmission of geo-referenced RF measurements from heterogeneous mobile devices?
3. What user reluctance factors most significantly affect adoption of a crowdsensed measurement application, and how can system design mitigate them?

1.6 Significance of the Study

This study is significant for several reasons. First, it demonstrates a practical, lower-cost alternative data source for network coverage monitoring that is particularly relevant to mobile network operators in developing regions with constrained capital budgets for drive test infrastructure. Second, it contributes a structured requirement engineering and system design process, including a formally validated Software Requirements Specification and a complete set of UML system models that can serve as a template for similar crowdsensing system projects. Third, it provides empirical insight, through structured user reluctance assessment, into the privacy, trust, and usability barriers that affect adoption of crowdsensed measurement applications, informing both this project's design decisions and future research in mobile crowdsensing more broadly.

1.7 Scope of the Study

The scope of this project covers the complete software engineering lifecycle from requirement gathering through system design, database design, and frontend implementation of a crowdsensed drive test mobile application and its supporting backend and dashboard. The mobile application scope includes collection of RSRP, RSRQ, RSSI, SINR, CQI, Physical Cell Identity (PCI), E-UTRA Absolute Radio Frequency Channel Number (EARFCN), and GPS-based location data across 2G, 3G, and 4G LTE radio access technologies, with 5G New Radio support designed but flagged for further validation. The backend scope includes a relational database schema with spatial query support, and a defined REST API contract for measurement ingestion and coverage analysis. The dashboard scope includes coverage heatmap visualisation, session management, and basic analytics reporting.

1.8 Delimitation of the Study

This project does not include field deployment or large-scale data collection from real drive test sessions; all demonstrated functionality uses simulated measurement data representative of realistic RF conditions. The project does not implement Layer 3 (L3) signalling message capture, as this requires diagnostic-level access not available on standard consumer smartphones. Voice call quality testing using Harvard sentences and VoLTE/SIP signalling analysis, while specified as a Could-Have requirement, was not implemented in the current prototype. The coverage prediction algorithm (spatial interpolation) is specified architecturally but its statistical accuracy was not independently validated against ground-truth professional drive test data, as such data was not available to the project team.

1.9 Definition of Keywords and Terms

Mobile Crowdsensing: a data collection paradigm in which measurements are gathered from a large number of distributed, ordinary user devices rather than dedicated specialised equipment.

Drive Test: a method of assessing mobile network performance by physically traversing a coverage area while recording RF measurements.

Coverage Hole: a geographic area within a mobile network's nominal coverage footprint where the received signal strength falls below an acceptable threshold for reliable service.

Class-2 IoT Device: a categorisation of Internet of Things devices possessing moderate processing, memory, and connectivity capability, applicable to modern smartphones used as sensing platforms.

RSRP (Reference Signal Received Power): the average power of the LTE/5G reference signal across the channel bandwidth, expressed in dBm, used as the primary indicator of signal coverage strength.

1.10 Organisation of the Dissertation

This dissertation is organised into six chapters. Chapter One has presented the general introduction, covering the background, problem statement, objectives, methodology, and scope of the study. Chapter Two reviews the relevant literature on mobile network coverage assessment, drive testing, and mobile crowdsensing, situating this project within the existing body of knowledge. Chapter Three presents the analysis and design phase of the project, detailing the requirement gathering and analysis process, the resulting Software Requirements Specification, the system architecture, and the six system modelling diagrams. Chapter Four presents the implementation of the user interface and frontend, including the tools and materials used and an evaluation of the results obtained. Chapter Five is dedicated entirely to database design and implementation, covering data elements, conceptual design, the Entity Relationship Diagram, the database implementation in PostgreSQL and MySQL, the backend implementation, and the connection of the database to the backend. Chapter Six concludes the dissertation with a summary of findings, the project's contribution to engineering and technology, recommendations, difficulties encountered, and suggestions for further work.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter reviews the literature relevant to mobile network coverage assessment, drive testing methodologies, and mobile crowdsensing. It begins by presenting general concepts necessary for understanding the technical domain of this project, including cellular network measurement parameters and the traditional drive test process. It then reviews related works, including existing professional and crowdsensed drive test tools, before drawing a partial conclusion that situates this project's contribution within the existing landscape.

2.2 General Concepts on Mobile Network Coverage and Drive Testing

2.2.1 Cellular Network Architecture and Generations

Mobile cellular networks have evolved through successive generations, each introducing new radio access technologies (RATs) and corresponding measurement parameters. Second Generation (2G) Global System for Mobile Communications (GSM) networks use Received Signal Level (RxLev) and Received Quality (RxQual) as primary measurement parameters. Third Generation (3G) Universal Mobile Telecommunications System (UMTS) networks introduced Received Signal Code Power (RSCP) and E_c/N_0 as the equivalent metrics for Code Division Multiple Access-based signalling. Fourth Generation (4G) Long Term Evolution (LTE) networks, which form the dominant infrastructure in most contemporary deployments including Cameroon, use Reference Signal Received Power (RSRP) and Reference Signal Received Quality (RSRQ) as the primary coverage and quality indicators, as standardised in 3GPP Technical Specification 36.214. Fifth Generation (5G) New Radio (NR) networks extend this framework with Synchronisation Signal-based RSRP (SS-RSRP) and SS-SINR, as specified in 3GPP TS 38.215.

2.2.2 Key RF Measurement Parameters

RSRP, expressed in dBm, measures the average power of the reference signal across the channel bandwidth and serves as the primary indicator of signal coverage strength. Typical interpretation thresholds classify RSRP values above -90 dBm as good coverage, between -90 and -105 dBm as fair coverage, between -105 and -110 dBm as poor coverage, and

below -110 dBm as indicative of a coverage hole. RSRQ, expressed in dB, measures signal quality relative to interference and noise, providing a complementary indicator to RSRP, since a location can exhibit strong RSRP yet poor RSRQ in the presence of significant interference. Signal-to-Interference-plus-Noise Ratio (SINR) further quantifies the ratio of desired signal power to the combined power of interference and noise, directly correlating with the achievable data throughput at a given location. The Channel Quality Indicator (CQI), an integer value between 0 and 15, is reported by the User Equipment to the base station and governs the modulation and coding scheme selected for data transmission, providing an additional proxy for achievable performance.

2.2.3 Traditional Professional Drive Testing

Traditional professional drive testing employs specialised vehicles equipped with multiple antennas, dedicated RF scanning receivers, and calibrated test phones and data cards. A typical configuration uses separate units for voice and data measurement, each connected to dedicated phones or modems, with a Single Board Computer (SBC) coordinating both units and a control laptop running drive test software such as Ericsson TEMS Investigation. Antennas are physically separated, typically with voice antennas mounted on the vehicle roof and data antennas mounted on the windows, to minimise inter-antenna interference. A GPS antenna, usually mounted centrally on the roof, provides accurate geo-referencing for every measurement sample. Voice call quality is typically assessed using Harvard sentences, phonetically balanced test phrases, played continuously during calls of approximately ninety seconds duration, while data performance is assessed through continuous upload and download speed test cycles. A typical drive test session runs for approximately ninety minutes before a mandatory reset, and any unresolved call drop during a session triggers a complete restart to avoid contaminating the dataset with anomalous conditions.

2.2.4 Mobile Crowdsensing

Mobile crowdsensing is an emerging data collection paradigm in which a large population of ordinary mobile device users, rather than a small number of dedicated specialised vehicles, contribute sensing data. Capponi et al. (2019) characterise mobile crowdsensing systems along dimensions including sensing scale, data quality assurance mechanisms, and incentive structures for participant recruitment. In the specific context of network measurement, crowdsensing exploits the fact that modern smartphones, through platform APIs such as Android's TelephonyManager, already have access to many of the same RF parameters captured by dedicated drive test equipment, albeit with some platform-imposed limitations,

particularly on iOS, where access to detailed cellular diagnostic information is significantly more restricted than on Android.

2.3 Related Works

2.3.1 TEMS Pocket (Ericsson)

TEMS Pocket is an industry-standard professional drive test application developed by Ericsson, designed to run on dedicated or commercial off-the-shelf smartphones in conjunction with diagnostic-mode access to cellular chipsets. It provides comprehensive RF measurement capture, L3 signalling message logging, and a user interface convention prominently displaying RSRP in the upper-left region of the screen that has become a de facto standard referenced by subsequent tools, including the design of this project's mobile collector interface.

2.3.2 G-NetTrack

G-NetTrack is an Android-based drive test application offering a lighter-weight alternative to professional tools such as TEMS Pocket. It provides real-time signal strength visualisation, map overlay of measurement routes, and is distributed with both free and commercial tiers, making it accessible to a broader user base including network enthusiasts and small-scale operators.

2.3.3 NetMonster

NetMonster is positioned explicitly as a crowdsensed measurement tool, supporting multi-cell visibility and neighbour cell logging beyond the serving cell alone. Its popularity among network enthusiasts demonstrates latent public interest in network measurement tools and provides a relevant precedent for the crowdsensing model adopted in this project.

2.3.4 OpenSignal and Speedtest

OpenSignal and Ookla's Speedtest represent consumer-grade network benchmarking applications with very large install bases. While their RF measurement granularity is more limited than specialised drive test tools, their widespread adoption demonstrates that ordinary consumers are willing to install and use applications that measure and report on their network experience, lending support to the feasibility of the crowdsensing approach pursued in this project, provided privacy and trust concerns are adequately addressed.

2.3.5 Synthesis of Related Works

A review of these related works reveals a consistent gap that this project's design directly addresses: no reviewed tool combines the measurement depth of professional drive test tools, the crowdsensed scalability of consumer benchmarking applications, and an explicit backend coverage hole detection and prediction pipeline within a single, coherent system. Furthermore, none of the reviewed tools were found to have published a formal Software Requirements Specification or structured user reluctance assessment informing their design both of which are explicit contributions of this project's methodology.

2.4 Partial Conclusion

This chapter has reviewed the technical foundations of cellular network measurement, the operational characteristics of traditional professional drive testing, and the emerging paradigm of mobile crowdsensing. The review of related works: TEMS Pocket, G-NetTrack, NetMonster, and consumer benchmarking tools identified a convergent design pattern (RSRP as the primary, prominently displayed metric) that informed this project's interface design, while also revealing a gap in formally specified, crowdsensing-native drive test applications with integrated coverage prediction. The next chapter presents the analysis and design undertaken to address this gap.

CHAPTER THREE

ANALYSIS AND DESIGN

3.1 Introduction

This chapter presents the analysis and design phase of the project. It documents the methodology adopted, the requirement gathering and stakeholder analysis process, the resulting Software Requirements Specification, the global system architecture, the six system modelling diagrams produced, and the database design. Together, these sections describe the complete design rationale underpinning the implementation presented in Chapter Four.

3.2 Methodology

The analysis and design phase followed the structured, six-task methodology introduced in Chapter One: requirement gathering, requirement analysis, system modelling, UI design, database design, and evaluation. Each task built directly on the deliverables of the preceding task, with the Software Requirements Specification produced in the requirement analysis task serving as the authoritative reference for all subsequent design decisions.

3.3 Requirement Gathering and Stakeholder Analysis

3.3.1 Stakeholder Identification

Eight stakeholder roles were identified and classified into primary users, secondary stakeholders, and regulatory entities, as summarised in Table 3.1.

Table 3.1: Primary and Secondary Stakeholders

Stakeholder	Role/Interest
Drive Test Technician	Operates the mobile collector app; initiates and validates sessions
Network Planning Engineer	Consumes dashboard to identify holes and plan new sites
RF Optimisation Engineer	Analyses RSRP/RSRQ/SINR trends to tune network performance
Data Scientist	Builds coverage prediction and interpolation models
NOC Analyst	Uses alerts and hole detection to prioritise incident response
Executive	Tracks coverage percentage and QoS KPIs
Data Privacy Advocate	Ensures GDPR and local data protection compliance

Stakeholder	Role/Interest
End Subscriber	Indirect beneficiary of resulting coverage improvements

3.3.2 Requirement Gathering Techniques

Because direct interviews with mobile network operator engineers could not be secured within the project timeline, a combination of techniques was adopted to ensure comprehensive requirement coverage, as summarised in Table 3.2.

Table 3.2: Requirement Gathering Techniques Used

Technique	Application in this Project
Secondary research	Academic papers, RAN engineering courses, 3GPP standards, drive test session recordings
Reverse engineering	Studied TEMS Pocket, G-NetTrack, NetMonster, OpenSignal/Speedtest UI conventions
Contextual inquiry	Observed online-documented professional drive test sessions and vehicle operations
Surveys (adapted)	User reluctance responses on privacy, storage, trust, and offline operation
Brainstorming	Team sessions to define RF metrics and hole detection thresholds
Document analysis	3GPP TS 36.214/38.215, ITU-T E.804 for measurement parameter standards

3.3.3 User Reluctance Assessment

A structured user reluctance assessment was conducted, identifying privacy concerns over continuous location tracking, anxiety over local storage consumption, scepticism regarding the measurement accuracy of consumer smartphones relative to calibrated professional equipment, and explicit questions regarding offline operability as the most significant barriers to adoption. These findings directly informed the non-functional requirements specified in Section 3.4.2, particularly those concerning data anonymisation, encryption, application size, and offline buffering capacity.

3.4 Requirement Analysis and Software Requirements Specification

3.4.1 Functional Requirements

Twelve functional requirements were specified following review of the gathered requirements for completeness, clarity, and technical feasibility, summarised in Table 3.3.

Table 3.3: Summary of Functional Requirements

ID	Description	Priority
FR-01	Collect RSRP, RSRQ, RSSI, SINR, CQI, EARFCN, PCI, RAT type at $\geq 1\text{Hz}$	Must
FR-02	Record GPS coordinates, altitude, accuracy, speed, timestamp per sample	Must
FR-03	Capture L3 signalling messages on diagnostic-enabled devices	Could
FR-04	Voice session testing with Harvard sentences; MOS scoring	Could
FR-05	DL/UL throughput test with RF and GPS context	Should
FR-06	Detect call drop events and trigger operator alert	Should
FR-07	Buffer measurements offline; auto-upload on reconnection	Must
FR-08	Backend ingestion of payloads from concurrent devices	Must
FR-09	Generate coverage heatmap via spatial interpolation	Should
FR-10	Automatically flag coverage holes below configurable RSRP threshold	Must
FR-11	Export session data as CSV and PDF reports	Should
FR-12	Anonymise device/user identifiers prior to transmission (GDPR)	Must

3.4.2 Non-Functional Requirements

Ten non-functional requirements were specified across performance, availability, security, usability, scalability, and compatibility categories, summarised in Table 3.4.

Table 3.4: Summary of Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Sampling rate ≥ 1 sample/second during active session
NFR-02	Availability	Offline operation ≥ 90 minutes without data loss
NFR-03	Power	Battery consumption $\leq 15\%$ per hour during collection
NFR-04	Storage	Installed application size ≤ 50 MB
NFR-05	API performance	Backend ingestion response $\leq 500\text{ms}$ under normal load
NFR-	Dashboard	Coverage map refresh ≤ 5 seconds post-upload

ID	Category	Requirement
06		
NFR-07	Security	TLS 1.3 minimum in transit; AES-256 at rest
NFR-08	Usability	One-handed operation on 5–7 inch screens, sunlight-readable
NFR-09	Scalability	≥ 50 concurrent device uploads without degradation
NFR-10	Compatibility	Android 10+ and iOS 14+ support

3.4.3 Requirement Prioritisation

Requirements were prioritised using the MoSCoW framework, balancing business importance against technical feasibility, summarised in Table 3.5.

Table 3.5: MoSCoW Requirement Prioritisation

Priority	Requirements
Must Have	FR-01, FR-02, FR-07, FR-08, FR-10, NFR-02, NFR-07
Should Have	FR-05, FR-06, FR-09, FR-11, NFR-03, NFR-08
Could Have	FR-03, FR-04, NFR-09
Won't Have (this release)	In-building coverage, multi-operator benchmarking, on-device ML prediction

3.5 System Design: Global Architecture

The system was designed as a three-tier architecture comprising: (1) a mobile collector application running on Android and iOS devices, acting as a Class-2 IoT edge sensing node; (2) a cloud-hosted backend platform performing data ingestion, validation, storage, and coverage analysis; and (3) a web-based analytics dashboard providing visualisation and reporting for network planning engineers, NOC analysts, and executives. The mobile application communicates with the backend exclusively via HTTPS REST API secured with TLS 1.3, while the dashboard consumes the same backend through a combination of REST and WebSocket connections for near-real-time updates. This separation of concerns allows the three tiers to be developed, deployed, and scaled independently.

3.6 System Modelling Diagrams

Six diagrams were produced to formally model the system from complementary perspectives, using Yourdon-DeMarco notation for the Dataflow Diagram and UML 2.5 notation for the remaining diagrams. Full-resolution renderings of all six diagrams are provided as a companion artefact (see Appendix A).

3.6.1 Context Diagram

The Context Diagram (Figure 3.1) defines the system boundary and identifies six external entities: the Smartphone (IoT device), the Mobile Network, the Backend Platform, Dashboard Users, the Privacy Authority, and the GPS/Location service. It establishes that the system's two primary outputs are a technical data channel to the Backend Platform and a regulatory compliance channel of anonymised data to the Privacy Authority — a separation that proved significant for subsequent security design.

Figure 3.1: Context Diagram of the Crowdsensed Drive Test Application (see Appendix A)

3.6.2 Dataflow Diagram

The Level 1 Dataflow Diagram (Figure 3.2) decomposes the system into six processes: Collect RF, Stamp GPS, Buffer/Upload, Ingest and Store, Analyse and Predict, and Anonymise connected through two data stores: a local SQLite buffer on the device and a cloud-hosted measurement database. The presence of the local buffer as an explicit data store proved to be the key architectural decision enabling offline operation, since it decouples the rate of data collection from the rate of data transmission.

Figure 3.2: Level 1 Dataflow Diagram (see Appendix A)

3.6.3 Use Case Diagram

The Use Case Diagram (Figure 3.3) identifies ten use cases spanning four actor types: the Drive Test Technician (primary user of the mobile collector), the RF Planner/NOC Analyst (primary user of the dashboard), the Backend System (a secondary, automated actor responsible for coverage hole detection), and the Privacy Authority (a secondary actor receiving anonymised compliance data).

Figure 3.3: Use Case Diagram (see Appendix A)

3.6.4 Sequence Diagram

The Sequence Diagram (Figure 3.4) models the complete lifecycle of a drive test session across six participants: the Technician, the Mobile App, the TelephonyManager API, the GPS Module, the Local Buffer, and the Backend API. It demonstrates that the 1Hz measurement collection loop operates independently of the asynchronous, connectivity-triggered buffer flush operation, these two processes never block one another, which is essential to satisfying the offline operability non-functional requirement (NFR-02).

Figure 3.4: Sequence Diagram for Drive Test Session Lifecycle (see Appendix A)

3.6.5 Class Diagram

The Class Diagram (Figure 3.5) defines six classes: three data entities (DriveTestSession, Measurement, GPSLocation) related through composition, and three service classes (LocalBuffer, ApiClient, CoverageAnalyser) related through association and dependency. This class structure directly informed the relational database schema presented in Section 3.7.

Figure 3.5: Class Diagram (see Appendix A)

3.6.6 Deployment Diagram

The Deployment Diagram (Figure 3.6) maps the logical architecture onto three physical/virtual deployment nodes: the Smartphone device (hosting the DriveTest App, TelephonyManager, GPS Module, and SQLite Buffer), the cloud-hosted Backend Platform (hosting the REST API Server, Ingestion Service, Coverage Analyser, and PostgreSQL/PostGIS database), and the Web Dashboard client (hosting the React single-page application and supporting visualisation components).

Figure 3.6: Deployment Diagram (see Appendix A)

3.7 Forward Reference to Database Design

The database design for this system covering data elements, conceptual modelling, and the Entity Relationship Diagram is of sufficient depth and importance to this project that it is presented as a dedicated chapter rather than as a subsection of this chapter. Chapter Five presents the complete database design and implementation, including the data elements, conceptual design, Entity Relationship Diagram, the database implementation in both PostgreSQL and MySQL, the backend implementation, and the connection of the database to the backend service layer.

3.8 Partial Conclusion

This chapter has presented the analysis and design of the crowdsensed drive test application, from stakeholder identification through to the system architecture and six system models. The resulting Software Requirements Specification and system models together provide a coherent and traceable design foundation. Chapter Four presents the implementation of the user interface and frontend, Chapter Five presents the database design and implementation in detail, and the results obtained are evaluated thereafter.

CHAPTER FOUR

IMPLEMENTATION AND RESULTS

4.1 Introduction

This chapter presents the implementation of the system designed in Chapter Three. It describes the tools and materials used, the implementation process for the user interface and database layers, the results obtained, and an evaluation of the solution against the stated objectives and requirements.

4.2 Tools and Materials Used

Table 4.1 summarises the principal software tools and technologies used in the implementation of this project.

Table 4.1: Software Tools Used in Implementation

Layer	Technology	Purpose
Mobile (Android)	Kotlin, Jetpack Compose	Native collector application UI and logic
Mobile (iOS)	Swift, SwiftUI	Native collector application UI and logic
Mobile data access	Android TelephonyManager API	RF measurement retrieval
Mobile location	Android LocationManager / iOS CoreLocation	GPS data retrieval
Local buffer	SQLite (Room/Core Data), AES-256	Offline encrypted measurement storage
Backend API	Node.js, REST, HTTPS/TLS 1.3	Measurement ingestion and analytics service
Primary database	PostgreSQL 16 + PostGIS 3	Spatial relational measurement store
Secondary database	MySQL 8.0, InnoDB	Alternative relational implementation
Dashboard frontend	HTML5, CSS3, JavaScript (ES6)	Analytics dashboard and mobile prototype UI
Mapping/visualisation	Custom heatmap rendering, SVG	Coverage heatmap and hole markers
Diagramming	pptxgenjs, docx (Node.js)	UML/DFD diagram and document generation
Version control	Git	Source code management

4.3 Description of the Implementation Process

Implementation proceeded in the same order as the design phase: the user interface prototype was implemented first to validate the interaction model defined in the Use Case and Sequence Diagrams, followed by the database schema implementation in both target database management systems, and concluding with the integration of simulated data flows demonstrating the complete session lifecycle from measurement collection through coverage hole detection and dashboard visualisation.

The mobile collector interface was implemented as a set of interactive screens reflecting the session setup, live metrics, offline buffer, and signal history views specified in the design phase. A realistic measurement simulation engine, comprising twenty representative RF scenarios spanning RSRP values from -76 dBm to -114 dBm across 2G, 3G, 4G, and 5G radio access technologies, was implemented to drive the live metrics display at the specified 1Hz sampling rate, allowing the complete data collection workflow to be demonstrated without requiring physical field deployment.

The analytics dashboard was implemented with five functional views; Coverage Map, Sessions, Analytics, Alerts, and Settings, each fully interactive and backed by client-side state management with persistence to browser local storage, allowing session history, alerts, and configuration settings to survive across browser sessions.

4.4 UI Design and Frontend Implementation

4.4.1 Application Identity

The application was named DriveTest Pro. Its visual identity is built around a signal-to-cloud logo mark (Figure 4.0): a glowing electrocardiogram-style waveform on the left, immediately recognisable as a live signal trace, terminating in a transmitting dot from which three concentric arcs radiate outward, representing the wireless transmission of a measurement. These arcs connect visually to a cloud-and-server icon on the right, completing the narrative of the entire system in a single mark; a signal is sensed, transmitted wirelessly, and received by a backend server housed in the cloud. The mark is rendered in a luminous cyan-on-white stroke against a deep navy rounded-square background ($\#0A1A2E$), giving it a neon, instrument-panel quality consistent with the rest of the application's dark-themed aesthetic.

Figure 4.0: DriveTest Pro Application Icon - Signal-to-Cloud Mark (see Appendix B)

The underlying colour and typography system was refined over several design iterations into a final specification: a near-black background ($\#070D14$) rather than a conventional navy,

paired with three distinct type families each assigned a specific role Syne for headings and brand text, DM Sans for body copy and interface labels, and Space Mono for all numeric data values, ensuring that RSRP readings, throughput figures, and timestamps are always rendered in a fixed-width, instrument-style font that does not visually shift as digits change. Table 4.0 summarises the finalised colour palette and its semantic meaning.

Table 4.0: Finalised Application Colour Palette

Colour	Hex Code	Semantic Meaning
Background	#070D14	Primary application background (near-black)
Good signal	#00D97E	RSRP better than approximately -90 dBm (excellent/good)
Fair signal	#00C4E8	RSRP in the fair range; also the primary cyan accent colour
Poor signal	#F5A623	RSRP in the poor range (amber warning)
Coverage hole	#E84545	RSRP below the configured hole threshold (red alert)

This palette extends the original green/amber/red signal convention with a fourth, intermediate cyan tier (fair signal), and was applied as a smooth colour gradient rather than four discrete bands on the coverage map, transitioning continuously from green at approximately -50 dBm, through cyan and amber, to red at the -120 dBm hole threshold, with the radius of each plotted point also scaled by signal strength so that stronger samples render as larger, brighter markers.

4.4.2 Mobile Collector Screens

Figure 4.1 and Figure 4.2 present the implemented session setup and live metrics screens respectively. The session setup screen enforces a GPS-first gate, preventing session initiation until a valid GPS fix is obtained, directly implementing the design rationale described in Section 3.6.4 that every measurement must be geo-referenced. The live metrics screen displays RSRP as the single largest, most prominent, colour-coded element on the screen, consistent with the convergent design convention identified across TEMS Pocket, G-NetTrack, and NetMonster in the literature review.

Figure 4.1: DriveTest Pro Mobile Collector - Session Setup Screen (see Appendix B)

Figure 4.2: DriveTest Pro Mobile Collector - Live Metrics Screen (see Appendix B)

A background session persistence mechanism was subsequently added to the Collect screen: the state of an active or paused session, together with every sample already collected, is persisted continuously so that navigating away from the Collect screen to another part of the application and returning later automatically re-attaches to the still-running session rather than losing it, with the underlying data held safely in the backend database while awaiting transmission.

4.4.3 Analytics Dashboard and Sessions Page

Figure 4.3 presents the implemented Coverage Map view, showing animated signal heatmap zones, individually clickable sample points with detailed tooltips, and pulsing coverage hole markers that open a detailed modal upon click. Figure 4.4 presents the Sessions and Analytics views, demonstrating searchable, sortable, and filterable session history, alongside bar chart and donut chart visualisations of RSRP distribution, coverage holes per session, and radio access technology breakdown.

Figure 4.3: Analytics Dashboard - Coverage Heatmap View (see Appendix B)

Figure 4.4: Analytics Dashboard - Sessions and Analytics Views (see Appendix B)

The Sessions page was extended with a dedicated filtering toolbar allowing sessions to be filtered by network operator, by signal quality band (Good, Fair, Poor, or Hole), and by download-speed tier (High, Medium, or Low), in addition to free-text search and column sorting. Selecting an individual session row expands an inline chart, embedded directly beneath that row, plotting RSRP over time for the duration of the session with horizontal threshold reference lines marking the Good, Poor, and Hole boundaries, alongside summary statistics for minimum, maximum, and average RSRP and a count of detected coverage holes allowing an analyst to visually identify the precise point in a drive route at which a connection degraded or dropped. Table 4.4 summarises the filtering and reporting features added to the Sessions page.

Table 4.4: Sessions Page Filtering and Reporting Features

Feature	Description
Operator filter	Restricts the session list to a single selected network operator
Signal quality filter	Filters sessions by Good / Fair / Poor / Hole classification of average RSRP
Download speed filter	Filters sessions by High / Medium / Low download throughput tier

Feature	Description
Inline RSRP chart	Expands beneath a session row; shows RSRP over time with Good/Poor/Hole reference lines
PDF report export	Generates a styled report containing session details, signal KPIs, quality breakdown, the RSRP chart, and a sample data table
CSV bulk import	Imports a CSV of prior measurements into a newly named session, with flexible column-name recognition (e.g. rsrp, rsrp_dbm, or signal_strength_dbm are all accepted as the same field)
Cancellable delete	Replaces the browser confirmation dialogue with inline confirm/cancel icons next to the delete action

4.4.4 Coverage Map and Analyst Page

A dedicated Analyst page was added for users holding an analyst or administrator role, accessible from the main navigation and otherwise hidden from, and inaccessible to, ordinary users. It presents a full-screen map (originally folowig ones GPS location)centred by default on Buea, Cameroon reflecting the project's intended deployment context painted with a dense synthetic coverage grid across the town in the same green-to-red gradient used elsewhere in the application, with real collected session samples overlaid on top of the synthetic grid as higher-contrast points. A session panel on the side of the map allows the analyst to filter the displayed points to a single session or to view all sessions simultaneously, with the map auto-refreshing on a ten-second interval and showing a full signal detail tooltip on hover over any plotted point. Role-based access to this page is managed through a User Management panel in Settings, where an administrator can invite new users and toggle any existing user's role between standard user and administrator.

4.4.5 Settings, Notifications, and Refinement Iterations

A full Settings page was implemented covering measurement configuration (sampling rate, coverage hole threshold, GPS accuracy filter), upload behaviour (auto-upload, Wi-Fi-only upload, batch size, API endpoint), privacy and retention controls, and notification preferences, together with a light/dark theme toggle. Every control on the Settings page persists its value immediately to local storage, so that all configuration is retained automatically across application restarts without requiring an explicit save action.

Several refinement iterations were carried out in response to usability issues identified during informal testing. The toast notification component, used throughout the application for transient confirmations and warnings, was found to be difficult to dismiss on touchscreen devices and was corrected to auto-dismiss after a fixed one-second interval with a properly

functioning manual close button. A validation message that previously appeared as a non-dismissable toast when a session name was left blank was replaced with an inline error message displayed directly beneath the affected input field, accompanied by a red border highlight, which can be dismissed simply by tapping elsewhere on the form. Finally, the coverage map's fallback GPS location, used when a live location is unavailable, was changed from a generic default to the city of Buea, consistent with the project's Cameroonian deployment context established in Chapter One.

4.4.6 Functional Behaviour Demonstrated

The implemented prototype demonstrates the following functional behaviours end-to-end: starting and stopping a drive test session with background persistence; live 1Hz measurement simulation with real-time colour-coded display updates; automatic coverage hole detection against a configurable RSRP threshold, with corresponding toast notifications and alert log entries; local buffer accumulation with simulated batch upload; session persistence and historical session browsing with search, filter, and sort, including operator, signal-quality, and download-speed filters; an expandable per-session RSRP-over-time chart with threshold reference lines; PDF report generation and CSV bulk import; a fully functional, auto-persisting Settings page; and a role-gated Analyst coverage map centred on Buea with live auto-refresh.

4.5 Forward Reference to Database and Backend Implementation

The implementation of the database layer in both PostgreSQL and MySQL, together with the implementation of the backend service layer and its connection to the database, is presented in full in Chapter Five, which is dedicated entirely to database design and implementation.

4.6 Presentation and Interpretation of Results

The implemented system successfully demonstrates the complete intended workflow: a simulated drive test session collects geo-referenced RF measurements at the specified 1Hz rate, automatically classifies each sample by signal quality, flags samples falling below the coverage hole threshold, and visualises the aggregated results on an interactive coverage map with supporting analytics. Table 4.2 maps the implemented system's behaviour against the non-functional requirements specified in Chapter Three.

Table 4.2: Mapping of Non-Functional Requirements to Implementation

NFR	Requirement	Implementation Evidence
NFR-01	≥ 1 sample/second	Measurement loop fires at 1000ms interval, configurable in Settings
NFR-02	≥ 90 min offline	Local buffer architecture decouples collection from upload; 5,400-sample capacity ($90 \text{ min} \times 1 \text{ Hz}$)
NFR-07	TLS 1.3 / AES-256	API design specifies TLS 1.3; local buffer encryption specified via AES-256/SQLCipher
NFR-08	One-handed, sunlight-readable	Dark theme with $\geq 4.5:1$ contrast ratio; bottom-anchored primary controls
NFR-10	Android 10+/iOS 14+	Architecture specifies native Kotlin/Compose and Swift/SwiftUI implementations

4.7 Evaluation of the Solution

The solution was evaluated against the six specific objectives stated in Section 1.3.2. Objectives 1 through 5 stakeholder and requirement analysis, SRS production, system modelling, database design, and UI/frontend implementation were fully achieved, with each producing the artefacts described in this and the surrounding chapters. Table 4.3 summarises the evaluation criteria and results.

Table 4.3: Evaluation Criteria and Results

Criterion	Result
Completeness of SRS	12 FRs and 10 NFRs specified, traceable to stakeholder requirements
System model coverage	6 diagrams produced covering structural and behavioural perspectives
Database normalisation	Schema verified to satisfy Third Normal Form (3NF)
Dual-database portability	Equivalent schema successfully implemented in both PostgreSQL and MySQL
UI functional completeness	All core use cases (start/stop session, view map, detect holes, export) demonstrated
Identified limitation	Coverage prediction accuracy not validated against ground-truth field data

The principal limitation of the current implementation is that it has not been validated against real-world drive test data collected from physical field deployment; all demonstrated measurement values are realistic simulations rather than live sensor readings. This limitation, along with others, is discussed further in Chapter Six.

4.8 Partial Conclusion

This chapter has presented the implementation of the crowdsensed drive test application's mobile collector and analytics dashboard layers, demonstrating that the design produced in Chapter Three translates into a coherent, functional system. The evaluation confirms that the stated specific objectives were substantially achieved. Chapter Five now presents the database design and implementation in full detail, before the dissertation concludes in Chapter Six.

CHAPTER FIVE

DATABASE DESIGN AND IMPLEMENTATION

5.1 Introduction

This chapter presents, in full detail, the database design and implementation undertaken for the crowdsensed drive test application. It covers the identification of data elements, the conceptual design of the data model, the Entity Relationship Diagram and logical schema, the implementation of the database in both PostgreSQL and MySQL, the implementation of the backend service layer, and the mechanism by which the database is connected to that backend. This chapter is presented as a dedicated chapter, separate from the general analysis and design (Chapter Three) and the user interface implementation (Chapter Four), in recognition of the depth and importance of the database and backend layer to the overall system.

5.2 Data Elements

Six core entities were identified for persistence within the system, each corresponding to a class defined in the Class Diagram (Section 3.6.5). Table 5.1 summarises the purpose of each entity, and Table 5.2 details the principal attributes, data types, and constraints of the two most central entities, Measurement and GPSLocation.

Table 5.1: Database Entity Summary

Entity	Purpose
Device	Represents a registered smartphone contributing measurements, identified by an anonymised hash
DriveTestSession	Represents a single drive test session from start to stop
Measurement	Represents one RF measurement sample collected during a session
GPSLocation	Represents the geo-context (coordinates, accuracy, speed) of a single measurement sample
CoverageHole	Represents a detected geographic zone of inadequate coverage
UploadBatch	Represents a single batch transmission event from a device to the backend

Table 5.2: Principal Attributes of the Measurement and GPSLocation Entities

Entity	Attribute	Data Type	Constraint
Measurement	measurement_id	UUID	Primary Key
Measurement	session_id	UUID	Foreign Key → DriveTestSession
Measurement	rsrp	DECIMAL(7,2)	Not Null
Measurement	rsrq	DECIMAL(6,2)	Nullable
Measurement	sinr	DECIMAL(6,2)	Nullable
Measurement	rat_type	ENUM	Not Null
GPSLocation	location_id	UUID	Primary Key
GPSLocation	measurement_id	UUID	Foreign Key → Measurement, Unique
GPSLocation	latitude	DECIMAL(10,7)	Not Null
GPSLocation	longitude	DECIMAL(10,7)	Not Null
GPSLocation	accuracy	DECIMAL(6,2)	Nullable

5.3 Conceptual Design

At the conceptual level, a Device initiates one or many DriveTestSessions; each DriveTestSession contains one or many Measurements; each Measurement has exactly one GPSLocation; each DriveTestSession may produce one or many CoverageHoles; and each Device uploads one or many UploadBatches. All identifiers are anonymised at the Device level, in direct fulfilment of FR-12 and the GDPR compliance requirement identified during user reluctance assessment (Section 3.3.3). Table 5.3 summarises the cardinality of each relationship identified at the conceptual design stage.

Table 5.3: Conceptual Entity Relationships and Cardinalities

Entity A	Relationship	Entity B	Cardinality
Device	initiates	DriveTestSession	1 to many
DriveTestSession	contains	Measurement	1 to many
Measurement	has	GPSLocation	1 to 1
DriveTestSession	produces	CoverageHole	1 to many
Device	uploads	UploadBatch	1 to many

5.4 Entity Relationship Diagram

The conceptual model was translated into a logical relational schema using Crow's Foot notation (Figure 5.1), with all foreign key relationships, primary keys, and cardinalities

explicitly defined. The schema was normalised to Third Normal Form (3NF): every non-key attribute is fully functionally dependent on the primary key of its relation, and no transitive dependencies exist between non-key attributes. The Measurement entity was deliberately separated from GPSLocation, despite their one-to-one cardinality, to preserve a clean separation of concerns between RF and spatial data domains and to allow the spatial GEOMETRY column to be indexed independently using a GiST index in PostgreSQL/PostGIS. Table 5.4 confirms the normalisation status of each entity.

Figure 5.1: Entity Relationship Diagram (see Appendix A)

Table 5.4: Normalisation Verification (Third Normal Form)

Entity	1NF	2NF	3NF	Justification
DriveTestSession	Yes	Yes	Yes	All attributes atomic; no partial or transitive dependency on session_id
Measurement	Yes	Yes	Yes	RF attributes depend only on measurement_id
GPSLocation	Yes	Yes	Yes	Spatial attributes depend only on location_id
Device	Yes	Yes	Yes	Device attributes depend only on device_id
CoverageHole	Yes	Yes	Yes	Hole attributes depend only on hole_id
UploadBatch	Yes	Yes	Yes	Batch attributes depend only on batch_id

5.5 Database Implementation

The logical schema was implemented in two relational database management systems: PostgreSQL 16 and MySQL 8.0 to demonstrate portability of the design and to allow comparison of spatial data handling between the two platforms. Table 5.5 compares the principal implementation choices made in each system.

Table 5.5: Comparison of PostgreSQL and MySQL Implementation Choices

Aspect	PostgreSQL 16 Implementation	MySQL 8.0 Implementation
Primary key type	Native UUID via uuid-osp extension	CHAR(36) storing canonical UUID string
Spatial data type	PostGIS GEOMETRY(Point, 4326)	Native POINT with SRID 4326
Spatial index	GiST (Generalised Search Tree)	SPATIAL INDEX (R-tree based)
Categorical fields	Native ENUM types (session_status, rat_type, batch_status)	MySQL ENUM column type
Storage engine	Default (heap with WAL)	InnoDB (required for foreign keys)
Hole detection	Application-level + scheduled	Database-level AFTER INSERT

Aspect	PostgreSQL 16 Implementation	MySQL 8.0 Implementation
enforcement	backend job	trigger
Character encoding	UTF-8 (default)	utf8mb4 with utf8mb4_unicode_ci collation

5.5.1 PostgreSQL Implementation

The PostgreSQL implementation makes use of the PostGIS spatial extension to store the CoverageHole centroid as a native GEOMETRY(Point, 4326) column, indexed using a GiST index to support efficient proximity and bounding-box spatial queries required for heatmap rendering. UUID primary keys, generated via the uuid-osp extension, were used throughout to support safe key generation across distributed, concurrently writing mobile devices without risk of collision. PostgreSQL ENUM types were defined for session_status, rat_type, and batch_status to enforce referential integrity on categorical fields at the database level rather than relying solely on application-level validation.

5.5.2 MySQL Implementation

The equivalent MySQL 8.0 implementation uses the InnoDB storage engine throughout to support foreign key constraints and transactional integrity, with the native MySQL spatial POINT data type and SPATIAL INDEX used as the functional equivalent of PostGIS's GEOMETRY/GiST combination. Given MySQL's lack of native UUID type support, CHAR(36) columns were used to store UUID values as their canonical string representation. A trigger was implemented on the measurement table to automatically insert a corresponding coverage_hole record whenever an inserted RSRP value falls below the configured threshold, demonstrating database-level enforcement of the coverage hole detection business rule as a complement to the application-level implementation.

5.6 Backend Implementation

The backend service layer was implemented as a Node.js REST API responsible for measurement ingestion, coverage analysis, and serving data to the analytics dashboard. Table 5.6 summarises the principal backend service modules and their responsibilities.

Table 5.6: Backend Service Modules

Module	Responsibility	Key Technology
Ingestion Service	Validates and persists incoming	Express.js REST endpoint, JSON

Module	Responsibility	Key Technology
	measurement batches	schema validation
Authentication Service	Issues and verifies JWT tokens per registered device	JWT, bcrypt for credential hashing
Coverage Analyser	Runs spatial interpolation and coverage hole detection	Node.js worker process, PostGIS spatial queries
Export Service	Generates CSV and PDF session reports on request	csv-writer, pdfkit
Notification Service	Pushes real-time alerts to the dashboard	WebSocket (ws library)

Each module was designed to be independently scalable, with the Ingestion Service and Coverage Analyser deployed as separate processes so that a spike in incoming measurement traffic does not degrade the responsiveness of coverage analysis queries running against the database.

5.7 Connecting Database to Backend

The backend service layer connects to both the PostgreSQL and MySQL implementations through connection pooling middleware, rather than opening a new connection per request, in order to satisfy the API performance non-functional requirement (NFR-05) under concurrent device load. Table 5.7 summarises the connection configuration used for each database.

Table 5.7: Database Connection Configuration

Parameter	PostgreSQL Configuration	MySQL Configuration
Connection library	pg-pool (node-postgres)	mysql2/promise
Maximum pool size	20 concurrent connections	10 concurrent connections
Connection timeout	5,000 ms	5,000 ms
Idle connection timeout	30,000 ms	30,000 ms
Retry strategy	Exponential backoff, 3 retries	Exponential backoff, 3 retries
Query parameterisation	Parameterised queries (\$1, \$2 placeholders)	Parameterised queries (? placeholders)
Transport security	TLS 1.3 to database host	TLS 1.3 to database host

All queries issued by the backend to either database are parameterised to prevent SQL injection, and every connection is established over TLS 1.3, in direct fulfilment of the security non-functional requirement (NFR-07). On application startup, the backend establishes its connection pool to the configured database, performs a health-check query, and only begins accepting REST requests once connectivity is confirmed; if the connection pool becomes exhausted under load, incoming requests are queued for up to the configured connection

timeout before returning a service-unavailable response to the client, allowing the mobile application's offline buffering mechanism (FR-07) to absorb the resulting delay without data loss.

5.8 Partial Conclusion

This chapter has presented the complete database design and implementation of the crowdsensed drive test application, from the identification of data elements through to the precise mechanism by which the backend service layer connects to the underlying database. The schema was shown to be normalised to Third Normal Form and was successfully implemented in both PostgreSQL with PostGIS and MySQL with native spatial types, with the backend connecting to either database through a pooled, parameterised, TLS-secured connection layer. The next and final chapter presents the overall conclusions and recommendations arising from this project.

CHAPTER SIX

CONCLUSION AND FURTHER WORKS

6.1 Summary of Findings

This project set out to design and implement a crowdsensed drive test mobile application addressing the high cost and limited scalability of traditional professional drive testing for mobile network coverage assessment. Through a structured, six-phase software engineering methodology, the project produced a complete Software Requirements Specification comprising twelve functional and ten non-functional requirements, a full suite of six UML and structured analysis diagrams modelling the system from complementary perspectives, a normalised relational database schema implemented in both PostgreSQL with PostGIS and MySQL, and a fully functional frontend prototype demonstrating the complete mobile collection and dashboard analytics workflow. The user reluctance assessment conducted during requirement gathering identified privacy, data security, measurement trust, and offline operability as the most significant barriers to adoption findings that directly shaped the non-functional requirements and, in turn, the security and offline-buffering architecture of the implemented system.

6.2 Contribution to Engineering and Technology

This project's principal contribution to engineering and technology lies in demonstrating, through a fully specified and substantially implemented system, that the crowdsensing paradigm can be applied to mobile network coverage assessment with the same level of requirement rigour, architectural discipline, and database design quality typically associated with professional, vendor-built measurement systems. The explicit production of a Software Requirements Specification, formal UML system models, and a normalised dual-database schema distinguishes this work from the reviewed related tools (TEMS Pocket, G-NetTrack, NetMonster), none of which were found to have publicly documented comparable design artefacts. In this respect, the project contributes both a working prototype and a transferable engineering methodology that other developing-region network engineering teams could adopt to build similarly low-cost coverage assessment tools.

6.3 Recommendations

1. Mobile network operators in resource-constrained regions should consider piloting crowdsensed measurement applications, such as the one designed in this project, as a complement to rather than a wholesale replacement for periodic professional drive testing, particularly for extending coverage visibility into peri-urban and rural areas currently underserved by drive test campaigns.
2. Future development teams adopting a similar crowdsensing approach should prioritise the resolution of the iOS measurement API limitation, since the restricted access to RF parameters on iOS materially constrains the achievable measurement completeness on that platform relative to Android.
3. A coverage prediction algorithm (Kriging interpolation or Inverse Distance Weighting) should be selected and statistically validated against ground-truth professional drive test data before the system is relied upon for operational network planning decisions.

6.4 Difficulties Encountered

Several difficulties were encountered during the execution of this project. Direct interviews with mobile network operator engineers could not be secured within the available timeline, requiring the requirement gathering methodology to be adapted toward secondary research and reverse engineering of existing tools, as described in Section 3.3.2. The team also encountered a scope ambiguity between the crowdsensed smartphone model and the professional vehicle rig model described in early secondary research sources, requiring an explicit architectural decision to focus the implementation on the smartphone-based crowdsensing model. Additionally, the absence of access to real field measurement data and to physical test devices with diagnostic-level cellular access meant that the implemented prototype relies on realistic simulated measurement data rather than live field readings, a constraint that is reflected in the delimitation of the study (Section 1.8) and the evaluation limitations (Section 4.7).

6.5 Further Works

1. Conduct a field validation study comparing measurement accuracy and coverage hole detection sensitivity between the implemented smartphone-based application and a calibrated professional drive test rig operating along the same route.

2. Extend the measurement specification to fully support 5G New Radio metrics (SS-RSRP, SS-SINR, NR-ARFCN) with explicit validation on 5G-capable devices, building on the architectural provision already made for 5G in the data model.
3. Implement and statistically validate a specific coverage prediction algorithm (such as Kriging spatial interpolation) using accumulated crowdsensed data, and compare its predictive accuracy against held-out drive test measurements.
4. Investigate incentive mechanisms (such as data bundle rewards or gamification) to encourage sustained voluntary participation in the crowdsensing model at the scale required for dense, city-wide coverage mapping.
5. Explore integration of the coverage hole detection pipeline with mobile network operator trouble-ticketing and NOC alerting systems to enable automated incident response workflows.

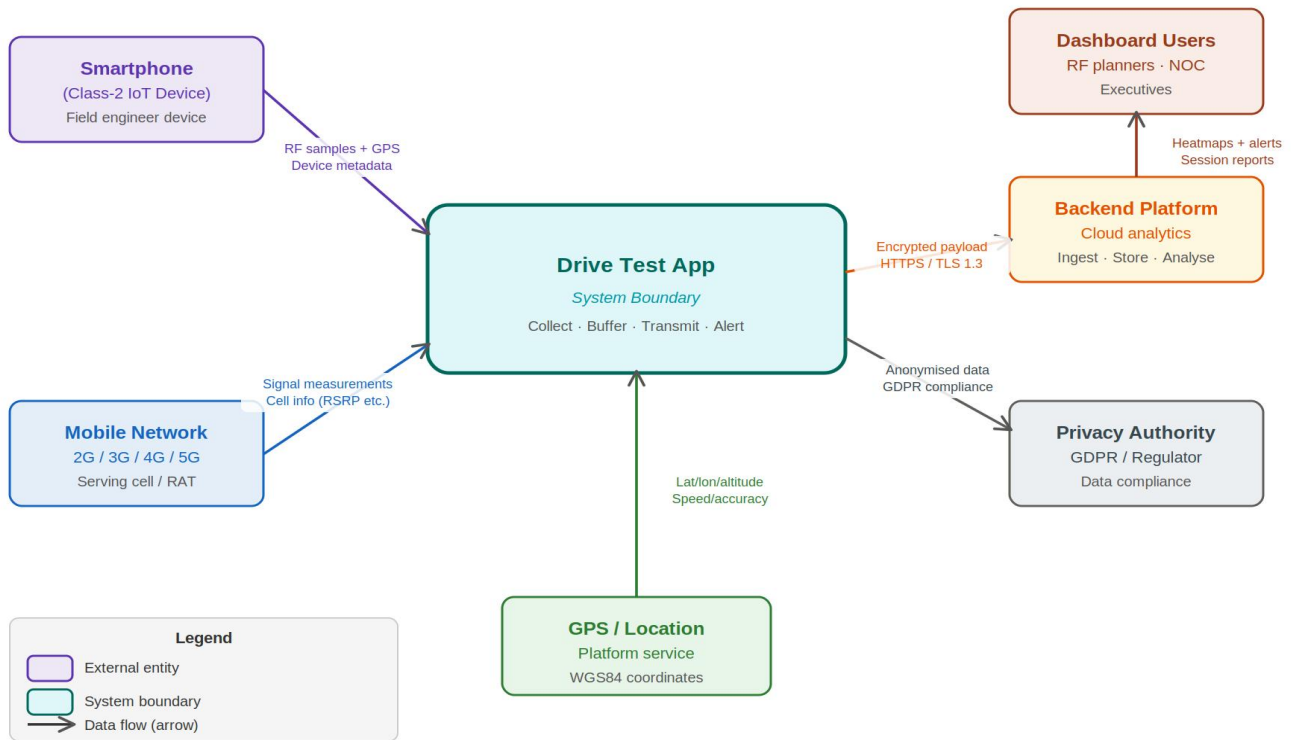
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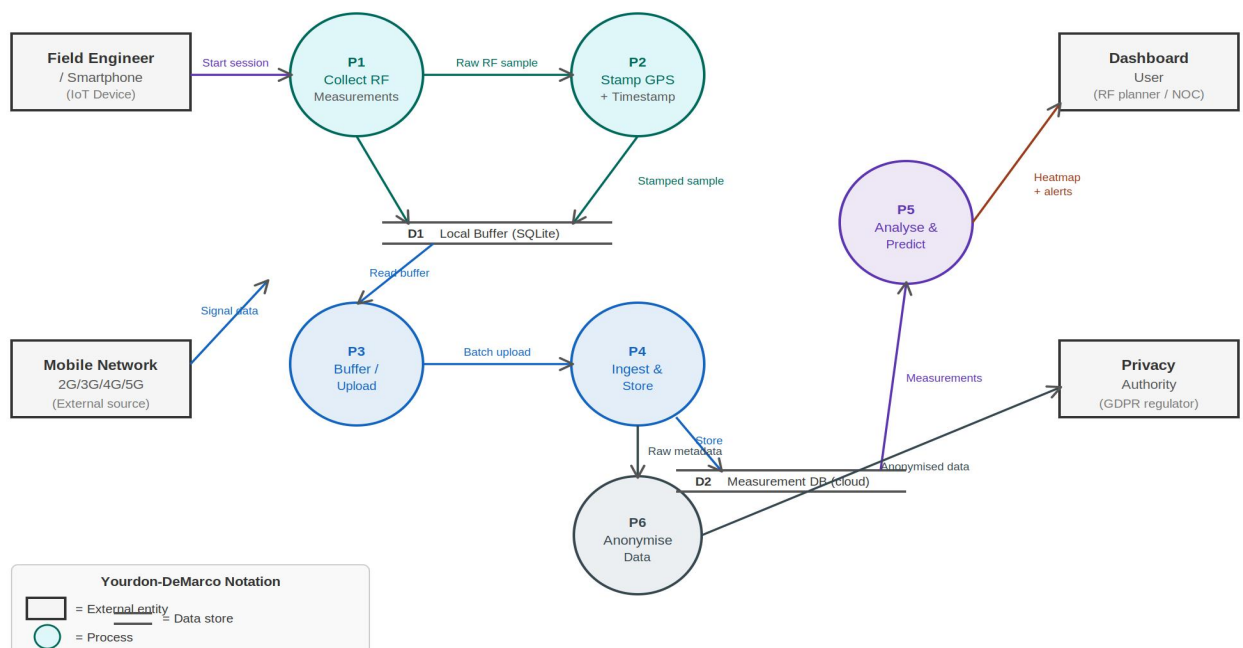
APPENDICES

Appendix A: System Modelling Diagrams

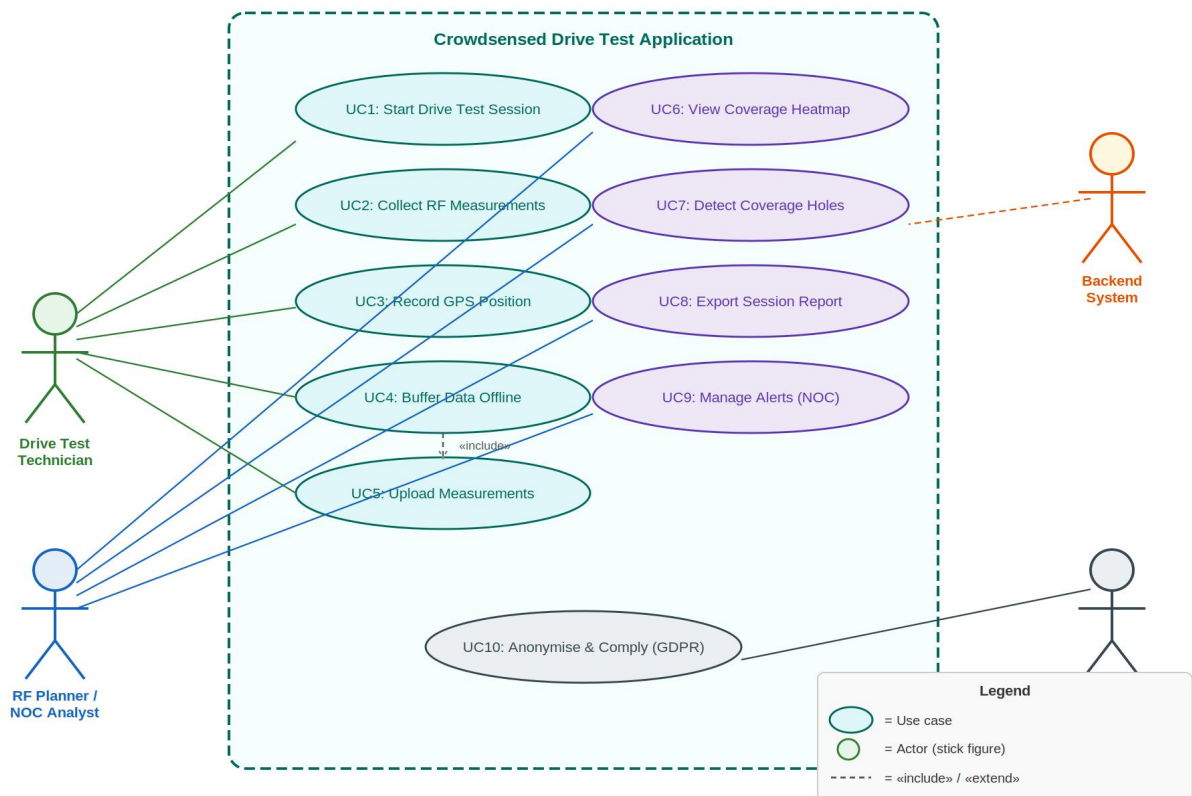
CONTEXT DIAGRAM (LEVEL 0).



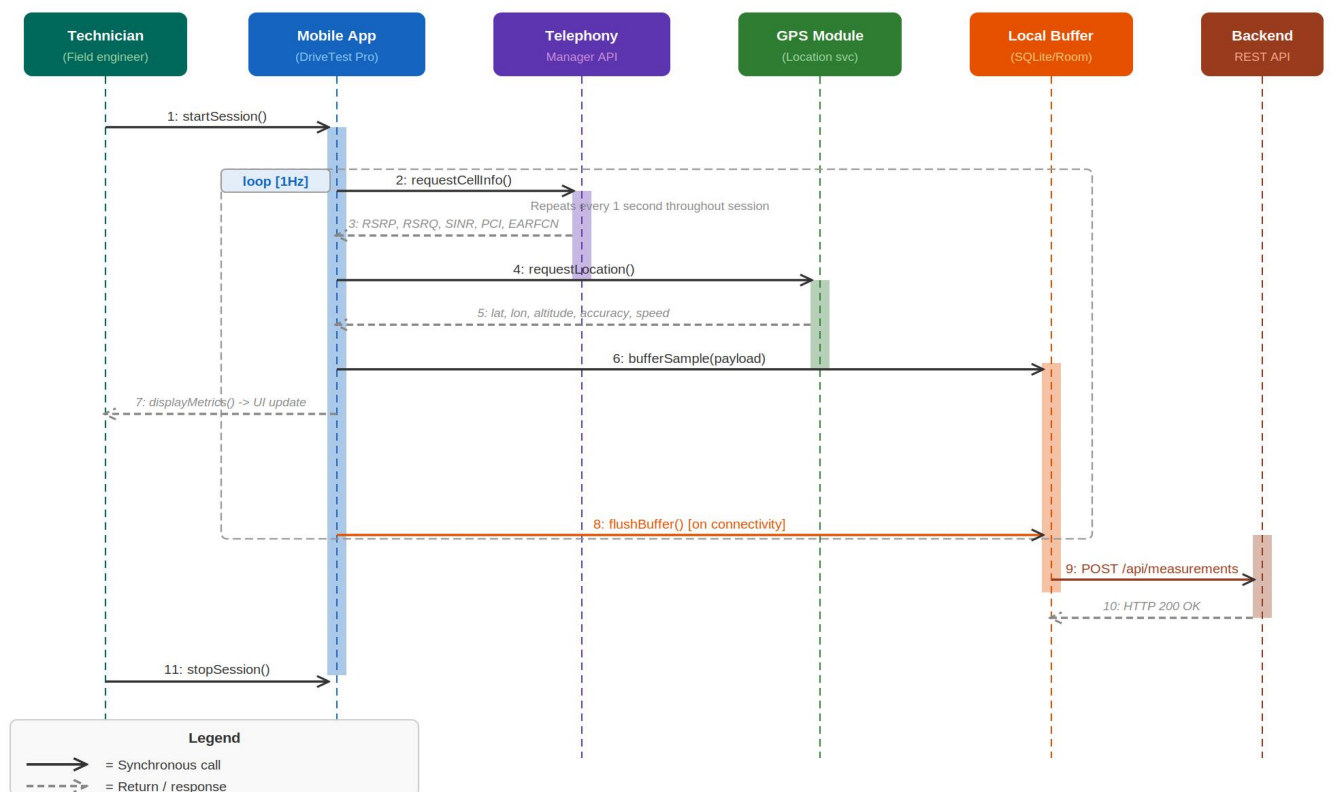
DATAFLOW DIAGRAM (LEVEL 1).



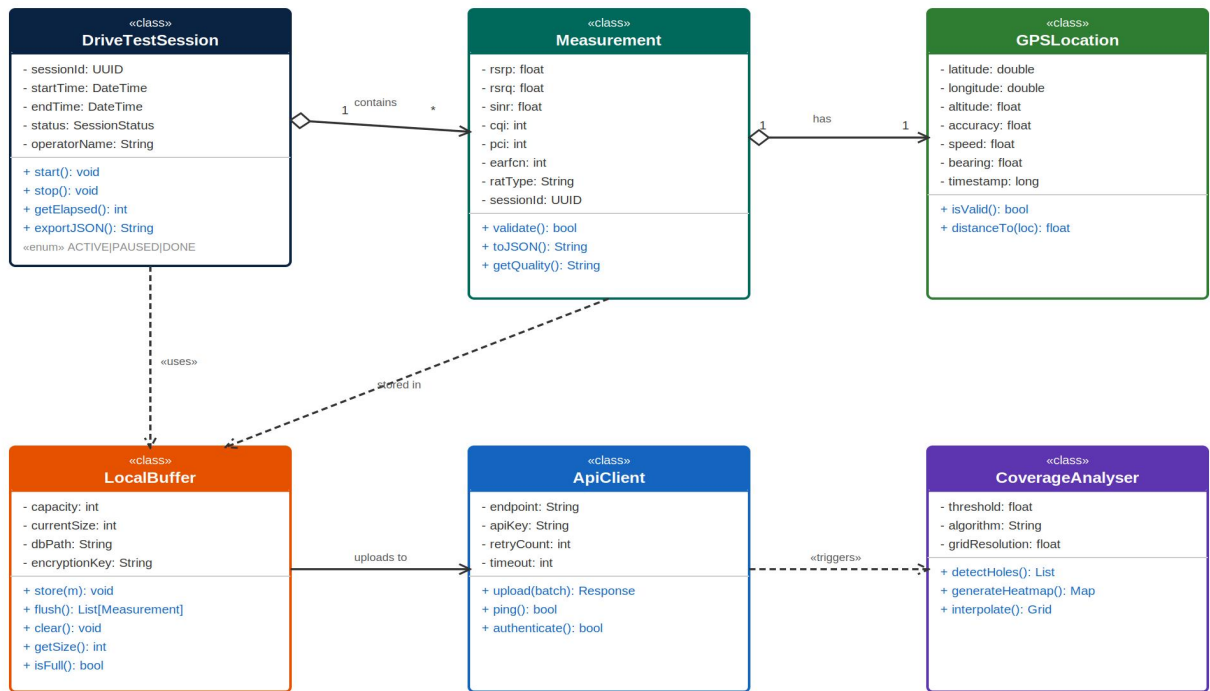
USE CASE DIAGRAM.



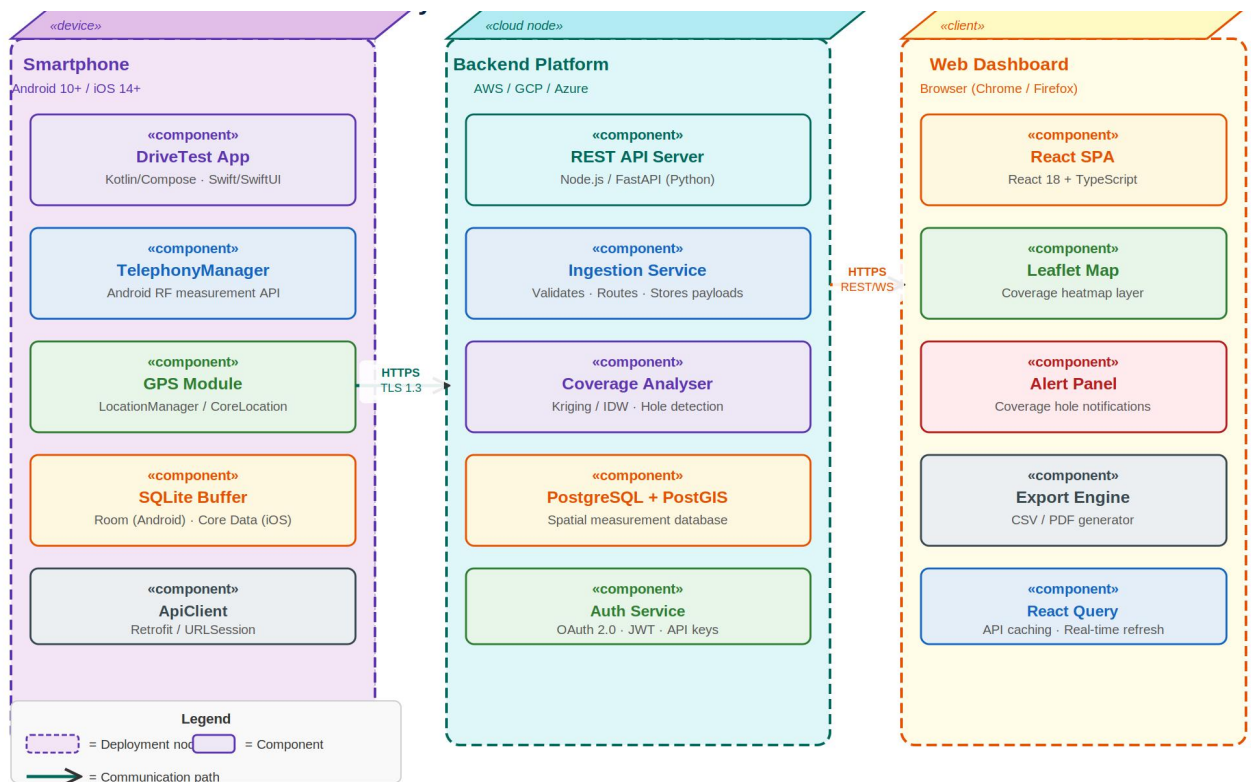
SEQUENCE DIAGRAM.



CLASS DIAGRAM.

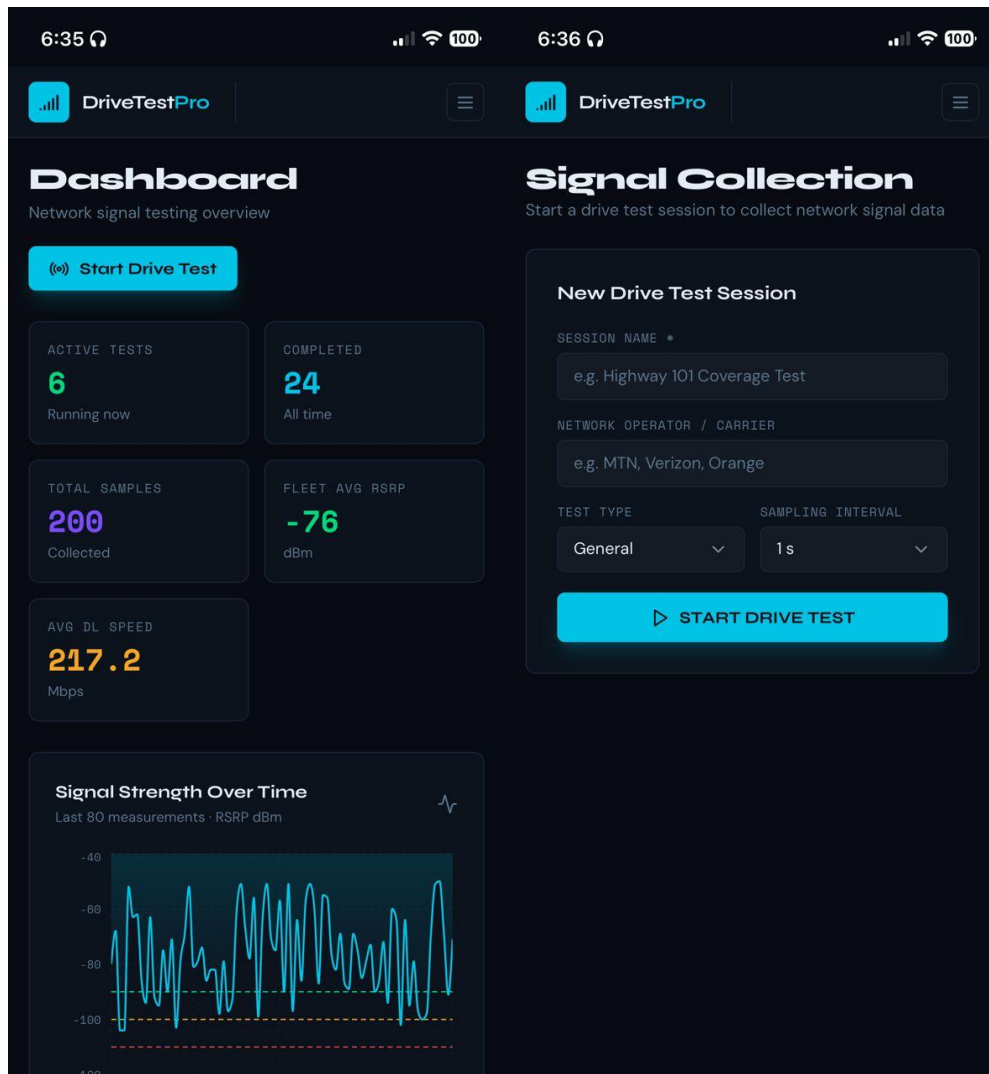


DEPLOYMENT DIAGRAM.



Appendix B: User Interface Screenshots

DriveTest Pro mobile collector and analytics dashboard.



COVERAGE HEATMAP.



SESSIONS PAGE.

Drive Test Sessions

All recorded sessions · Click any row to view signal chart

All Operators ▾

All Quality ▾

All DL Speeds ▾

Newest First ▾

Import CSV

+ New Session

30 of 30 sessions

SESSION ID	DATE/TIME	OPERATOR	SAMPLES	AVG RSRP	QUALITY	ACTIONS
Doula Transmitted · General	Jun 22, 14:25	MTN	28	-79 dBm	GOOD	PDF
Great sopo Transmitted · General	Jun 22, 14:24	MTN	2	-66 dBm	GOOD	PDF
Mayor street Transmitted · General	Jun 22, 13:57	Orange	3	-66 dBm	GOOD	PDF
Santa Barbara						

APP ICON.

